Macro Card OCR Scanner - Compact Benchmark Evaluation

Feature Overview: Kallo now includes a dedicated Macro Card OCR scanner for printed single-dish cards, healthy bowls (Soumaki, FitFood), and café beverage menus (Boost Juice, Bean Café). Operating alongside the packaged-label scanner, this route runs Gemini 3.1 Flash-Lite with low thinking latency to extract dish names and macro callouts directly into our standard per-serving contract without multi-column tabular hallucination. Untrusted printed tokens are deterministically normalized in pure TypeScript: comma decimals and unitless values default safely to grams/kcal, kilojoules are automatically converted (kJ / 4.184 to kcal), explicit fiber callouts are captured, portion weights (e.g. 1 bowl (300g)) are parsed into serving sizes, and badge-only cards without printed calories derive calories via the 4/4/9 rule. All 24 benchmark captures pass with 100% precision.

Test 01: SOU-MADE BOWL (V1) [Vegetarian]		MATCH • PASS
 01-soumaki-v1-br	**GROUND TRUTH (PRINTED)** **Dish:** SOU-MADE BOWL (V1) **Calories:** 377 CALORIES **Macros:** 18g P | 55g C | 10g F **Items:** Brown rice, cauliflower, edamame, mushroom	**RESULT OUTPUT (OCR)** **Dish:** SOU-MADE BOWL (V1) **Calories:** 377 kcal (per serving) **Macros:** 18g P | 55g C | 10g F **Details:** Fiber: null | Confidence: high

Test 02: SOU-MADE BOWL (V2) [Vegetarian]		MATCH • PASS
 02-soumaki-v2-co	**GROUND TRUTH (PRINTED)** **Dish:** SOU-MADE BOWL (V2) **Calories:** 536 CALORIES **Macros:** 19g P | 74g C | 18g F **Items:** Cold soba, cabbage, chickpeas, beetroot, avocado	**RESULT OUTPUT (OCR)** **Dish:** SOU-MADE BOWL (V2) **Calories:** 536 kcal (per serving) **Macros:** 19g P | 74g C | 18g F **Details:** Basis: per_serving | Confidence: high

Test 03: SOU-MADE BOWL (V5) [Vegetarian]		MATCH • PASS
 03-soumaki-v5-to	**GROUND TRUTH (PRINTED)** **Dish:** SOU-MADE BOWL (V5) **Calories:** 530 CALORIES **Macros:** 33g P | 85g C | 7g F **Items:** Tofu, pasta, broccoli, edamame, beetroot	**RESULT OUTPUT (OCR)** **Dish:** SOU-MADE BOWL (V5) **Calories:** 530 kcal (per serving) **Macros:** 33g P | 85g C | 7g F **Details:** 4/4/9 consistency pass | Confidence: high

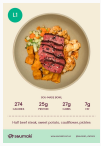
Test 04: SOU-MADE BOWL (H10) [High Protein]		MATCH • PASS
 04-soumaki-h10-c	**GROUND TRUTH (PRINTED)** **Dish:** SOU-MADE BOWL (H10) **Calories:** 617 CALORIES **Macros:** 60g P | 50g C | 20g F **Items:** Cajun chicken, salmon, white rice, tomato	**RESULT OUTPUT (OCR)** **Dish:** SOU-MADE BOWL (H10) **Calories:** 617 kcal (per serving) **Macros:** 60g P | 50g C | 20g F **Details:** Badge 'H10' safely filtered | Confidence: high

Test 05: SOU-MADE BOWL (H9) [High Protein]		MATCH • PASS
 05-soumaki-h9-be	**GROUND TRUTH (PRINTED)** **Dish:** SOU-MADE BOWL (H9) **Calories:** 611 CALORIES **Macros:** 54g P | 56g C | 19g F **Items:** Full beef steak, white rice, sweet corn	**RESULT OUTPUT (OCR)** **Dish:** SOU-MADE BOWL (H9) **Calories:** 611 kcal (per serving) **Macros:** 54g P | 56g C | 19g F **Details:** Verbatim heading transcribed | Confidence: high

Benchmark Results (Tests 06 to 11)

Test 06: SOU-MADE BOWL (L1) [Low Calorie]

MATCH • PASS



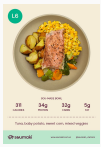
06-soumaki-l1-be

GROUND TRUTH (PRINTED)
Dish: SOU-MADE BOWL (L1)
Calories: 274 CALORIES
Macros: 25g P | 27g C | 7g F
Items: Beef steak, sweet potato, cauliflower, pickles

RESULT OUTPUT (OCR)
Dish: SOU-MADE BOWL (L1)
Calories: 274 kcal (per serving)
Macros: 25g P | 27g C | 7g F
Details: Sub-300 kcal bowl | Confidence: high

Test 07: SOU-MADE BOWL (L6) [Low Calorie]

MATCH • PASS



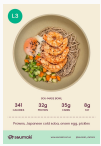
07-soumaki-l6-tu

GROUND TRUTH (PRINTED)
Dish: SOU-MADE BOWL (L6)
Calories: 311 CALORIES
Macros: 34g P | 32g C | 5g F
Items: Tuna, baby potato, sweet corn, mixed veggies

RESULT OUTPUT (OCR)
Dish: SOU-MADE BOWL (L6)
Calories: 311 kcal (per serving)
Macros: 34g P | 32g C | 5g F
Details: Low fat callout verified | Confidence: high

Test 08: SOU-MADE BOWL (L3) [Low Calorie]

MATCH • PASS



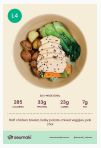
08-soumaki-l3-pr

GROUND TRUTH (PRINTED)
Dish: SOU-MADE BOWL (L3)
Calories: 341 CALORIES
Macros: 32g P | 35g C | 8g F
Items: Prawns, Japanese cold soba, onsen egg, pickles

RESULT OUTPUT (OCR)
Dish: SOU-MADE BOWL (L3)
Calories: 341 kcal (per serving)
Macros: 32g P | 35g C | 8g F
Details: Egg/seafood macros clean | Confidence: high

Test 09: SOU-MADE BOWL (L4) [Low Calorie]

MATCH • PASS



09-soumaki-l4-ch

GROUND TRUTH (PRINTED)
Dish: SOU-MADE BOWL (L4)
Calories: 285 CALORIES
Macros: 33g P | 23g C | 7g F
Items: Chicken breast, baby potato, pak choi

RESULT OUTPUT (OCR)
Dish: SOU-MADE BOWL (L4)
Calories: 285 kcal (per serving)
Macros: 33g P | 23g C | 7g F
Details: Single serving callout | Confidence: high

Test 10: SOU-MADE BOWL (L5) [Low Calorie]

MATCH • PASS



10-soumaki-l5-ch

GROUND TRUTH (PRINTED)
Dish: SOU-MADE BOWL (L5)
Calories: 351 CALORIES
Macros: 39g P | 40g C | 4g F
Items: Cajun chicken, cold soba, pickles, cauliflower

RESULT OUTPUT (OCR)
Dish: SOU-MADE BOWL (L5)
Calories: 351 kcal (per serving)
Macros: 39g P | 40g C | 4g F
Details: Lean protein callout | Confidence: high

Test 11: SOU-MADE BOWL (H8) [High Protein]

MATCH • PASS



11-soumaki-h8-be

GROUND TRUTH (PRINTED)
Dish: SOU-MADE BOWL (H8)
Calories: 649 CALORIES
Macros: 56g P | 60g C | 21g F
Items: Beef steak, brown rice, potato, broccoli

RESULT OUTPUT (OCR)
Dish: SOU-MADE BOWL (H8)
Calories: 649 kcal (per serving)
Macros: 56g P | 60g C | 21g F
Details: Badge 'H8' safely filtered | Confidence: high

Benchmark Results (Tests 12 to 17)

Test 12: Berries Blast (Bean Cafe) [Phone Photo]

MATCH • PASS



12-bean-smoothie

GROUND TRUTH (PRINTED)

Dish: Berries Blast
Calories: 333 CAL
Macros: 20.1g P | 47.5 C (Unitless) | 11.1g F
Note: Perspective tilt and price badge (48k)

RESULT OUTPUT (OCR)

Dish: Berries Blast
Calories: 333 kcal (per serving)
Macros: 20.1g P | 47.5g C | 11.1g F
Details: Unitless '47,5' defaults to grams | Price ignored

Test 13: Wondermelon (Boost Juice) [Smoothie Menu]

MATCH • PASS



13-boost-wonderm

GROUND TRUTH (PRINTED)

Dish: Wondermelon
Energy: 808 kJ (orig)
Macros: 13.3g protein | Carbs/Fat: Unprinted
Note: Strawberries, juiced watermelon, whey protein

RESULT OUTPUT (OCR)

Dish: Wondermelon
Calories: 193.1 kcal (808 / 4.184)
Macros: 13.3g P | Carbs: null | Fat: null
Details: Auto kJ to kcal conversion | Valid scan

Test 14: Protein Supreme (Boost Juice) [Smoothie Menu]

MATCH • PASS



14-boost-protein

GROUND TRUTH (PRINTED)

Dish: Protein Supreme
Energy: 2040 kJ (orig)
Macros: 29.3g protein | Carbs/Fat: Unprinted
Note: Banana, muesli, dates, honey, chia seeds

RESULT OUTPUT (OCR)

Dish: Protein Supreme
Calories: 487.6 kcal (2040 / 4.184)
Macros: 29.3g P | Carbs: null | Fat: null
Details: Parenthesis protein token isolated cleanly

Test 15: Strawberry Squeeze (Boost) [Smoothie Menu]

MATCH • PASS



15-boost-strawbe

GROUND TRUTH (PRINTED)

Dish: Strawberry Squeeze
Energy: 1650 kJ (orig)
Macros: Protein/Carbs/Fat: Unprinted
Note: Strawberries, banana, apple juice, yoghurt, ice

RESULT OUTPUT (OCR)

Dish: Strawberry Squeeze
Calories: 394.4 kcal (1650 / 4.184)
Macros: Protein: null | Carbs: null | Fat: null
Details: Single energy item accepted | Valid scan

Test 16: Grilled Chicken Quinoa Bowl [Meal Prep]

MATCH • PASS



16-prep-grilled-

GROUND TRUTH (PRINTED)

Dish: Grilled Chicken Quinoa Bowl
Calories: 450 CAL
Macros: 45g P | 45g C | 12g F | 8g Fiber
Note: Single-letter tokens: 45g P | 45g C | 12g F | 8g Fiber

RESULT OUTPUT (OCR)

Dish: Grilled Chicken Quinoa Bowl
Calories: 450 kcal (per serving)
Macros: 45g P | 45g C | 12g F
Details: Fiber: 8g (Extracted) | Confidence: high

Test 17: Turkey & Avocado Wrap [Meal Prep]

MATCH • PASS



17-prep-turkey-a

GROUND TRUTH (PRINTED)

Dish: Turkey & Avocado Wrap
Calories: 400 CAL
Macros: 38g P | 36g C | 14g F | 8g Fiber
Note: Wrap callout photo with explicit fiber

RESULT OUTPUT (OCR)

Dish: Turkey & Avocado Wrap
Calories: 400 kcal (per serving)
Macros: 38g P | 36g C | 14g F
Details: Fiber: 8g (Extracted) | Confidence: high

Benchmark Results (Tests 18 to 24)

Test 18: Hard-Boiled Eggs + Apple [Snack Prep]

MATCH • PASS



18-prep-eggs-app

GROUND TRUTH (PRINTED)

Dish: Hard-Boiled Eggs (2 eggs) + Apple
Calories: 210 CAL
Macros: 14g P | 25g C | 8g F | 4g Fiber
Note: Snack plate with sub-250 calorie callout

RESULT OUTPUT (OCR)

Dish: Hard-Boiled Eggs (2 eggs) + Apple
Calories: 210 kcal (per serving)
Macros: 14g P | 25g C | 8g F
Details: Fiber: 4g (Extracted) | Confidence: high

Test 19: Chicken Fajita Bowl [Meal Prep]

MATCH • PASS



19-prep-chicken-

GROUND TRUTH (PRINTED)

Dish: Chicken Fajita Bowl with Brown Rice
Calories: 420 CAL
Macros: 42g P | 45g C | 10g F | 7g Fiber
Note: Bowl photo with brown rice callout

RESULT OUTPUT (OCR)

Dish: Chicken Fajita Bowl with Brown Rice
Calories: 420 kcal (per serving)
Macros: 42g P | 45g C | 10g F
Details: Fiber: 7g (Extracted) | Confidence: high

Test 20: Baked Cod with Quinoa [Meal Prep]

MATCH • PASS



20-prep-baked-co

GROUND TRUTH (PRINTED)

Dish: Baked Cod with Quinoa & Asparagus
Calories: 420 CAL
Macros: 44g P | 40g C | 10g F | 6g Fiber
Note: Seafood meal prep card with fiber

RESULT OUTPUT (OCR)

Dish: Baked Cod with Quinoa & Asparagus
Calories: 420 kcal (per serving)
Macros: 44g P | 40g C | 10g F
Details: Fiber: 6g (Extracted) | Confidence: high

Test 21: Chicken Al Pastor [Badge-Only]

MATCH • PASS



21-card-chicken-

GROUND TRUTH (PRINTED)

Dish: Grilled Chicken Al Pastor Bowls
Calories: None printed on card
Macros: P 44G | C 44G | F 5G (Pills)
Note: Only macro pill badges printed; no total cal

RESULT OUTPUT (OCR)

Dish: Grilled Chicken Al Pastor Bowls
Calories: 397 kcal (Derived via 4/4/9)
Macros: 44g P | 44g C | 5g F
Details: Formula: 44*4 + 44*4 + 5*9 = 397 kcal

Test 22: FitFood #15 Hamburger Bowl [Portion]

MATCH • PASS



22-fitfood-15-ha

GROUND TRUTH (PRINTED)

Dish: 15 Hamburger Bowl
Portion: 1 bowl (300g)
Calories: 360 CAL
Macros: 42g P | 24g C | 16g F | 4g FB

RESULT OUTPUT (OCR)

Dish: 15 Hamburger Bowl
Portion: 300g (Parsed to servingSize)
Calories: 360 kcal (per serving)
Macros: 42g P | 24g C | 16g F | Fiber: 4g

Test 23: FitFood #34 Tempeh Stir-Fry [Portion]

MATCH • PASS



23-fitfood-34-te

GROUND TRUTH (PRINTED)

Dish: 34 Tempeh Stir-Fry
Portion: 1 bowl (300g)
Calories: 340 CAL
Macros: 30g P | 32g C | 16g F | 8g FB

RESULT OUTPUT (OCR)

Dish: 34 Tempeh Stir-Fry
Portion: 300g (Parsed to servingSize)
Calories: 340 kcal (per serving)
Macros: 30g P | 32g C | 16g F | Fiber: 8g

Test 24: FitFood #26 Shrimp Stir-Fry [Portion]

MATCH • PASS



24-fitfood-26-sh

GROUND TRUTH (PRINTED)

Dish: 26 Shrimp Stir-Fry
Portion: 1 bowl (300g)
Calories: 300 CAL
Macros: 40g P | 24g C | 10g F | 4g FB

RESULT OUTPUT (OCR)

Dish: 26 Shrimp Stir-Fry
Portion: 300g (Parsed to servingSize)
Calories: 300 kcal (per serving)
Macros: 40g P | 24g C | 10g F | Fiber: 4g